

IndoorPulse: Infrastructure-Free Indoor Positioning System

Client-Side Indoor Wayfinding via Sensor Fusion, Dynamic Graphing, and AR Overlays

Core Goal

Turn-by-turn indoor positioning on standard smartphones without external beacons, venue permissions, or manual CAD blueprints.

Indoor Positioning Challenges and Target Users

The Problem Space

Physical Barrier

GPS signals fail to penetrate concrete and steel building structures, causing complete indoor location failure.

Infrastructure Bottleneck

Commercial indoor systems require expensive dedicated beacon hardware, regular battery upkeep, and manual floor mapping.

Intended Users

1. Unmapped Venues and Cultural Sites

Visitors navigating multi-acre religious complexes, pilgrimage grounds, and dense informal markets lacking structural CAD blueprints, beacon infrastructure, or cellular GPS.

2. Academic and Healthcare Campuses

Students, staff, and visitors requiring precise room-level, multi-floor navigation across interconnected university departments, research labs, and hospital wings.

3. Accessibility and Step-Free Transit Users

Mobility-impaired and wheelchair users requiring automated route filtering that excludes stairwells in favor of verified ramps and elevators.

4. Crisis Evacuation and Life Safety (SOS Mode)

Building occupants during power outages, fire hazards, or low-visibility emergencies requiring immediate offline routing to the nearest verified emergency exit.

Functional Capabilities and Hardware Interfaces

Core Functions

Zero-Plan SLAM

Builds walkable corridor graphs as users move through unmapped spaces.

Real-Time Motion Tracking

Ingests 50Hz IMU motion data on-device to track steps and direction.

Particle Filter Bounding

Constrains user position inside hallways using 400 candidate particles.

Vertical Layer Switching

Detects floor changes using relative barometric air pressure changes.

Hardware Interfaces

Android

SensorManager (Accelerometer, Gyroscope, Magnetometer), Barometer API, WifiManager, ARCore.

iOS

CoreMotion (IMU), CMAltimeter, Magnetometer, ARKit Visual-Inertial Odometry.

Cloud API

HTTPS REST endpoints for publishing versioned deliverables and syncing map graphs.

Step Detection and Adaptive Stride Calculation

Step Detection Pipeline

01

Magnitude Computation

Combines 3-axis accelerometer data into an orientation-independent magnitude: $a(t) = \sqrt{a_x^2 + a_y^2 + a_z^2}$

02

Butterworth Bandpass Filter

Applies a 1Hz to 3Hz bandpass filter to capture human walking impacts while removing phone tilt and vehicle vibrations.

03

Peak Detection

Uses peak detection with a 300ms delay to prevent false double-counting.

Weinberg Stride Length Model

Formula: Stride Length = $k * (a_{\text{max}} - a_{\text{min}})^{(1/4)}$

Dynamically scales step length based on vertical bounce amplitude, adapting automatically to fast walking, slow walking, and stairs.

Heading Fusion

Merges gyroscope rotation data with the 3-axis magnetometer using a complementary filter to eliminate directional drift.

Probability Cloud Fusion and Wall Bounding

State Space

Tracks **400 candidate position guesses** (X, Y, Heading) initialized near the building entrance.

Convergence

Resamples the particle pool every step, collapsing the cloud into an accurate user coordinate.

Combined Weight = Wall Mask * Signal Match * Weak GPS * Anchor Proximity

Wall Mask

Instantly deletes particles that cross solid walls or room boundaries.

Signal Match

Compares local magnetic field anomalies or Wi-Fi signatures against stored floor data.

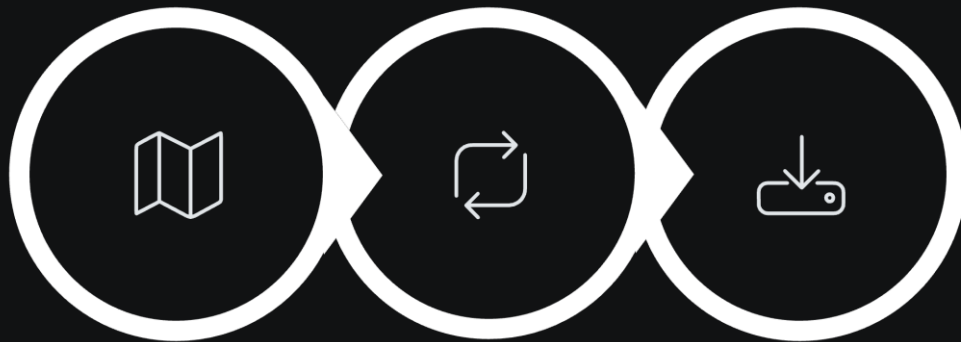
Weak GPS

Uses rough indoor GPS accuracy bounds to keep the particle cloud from drifting to wrong building wings.

Anchor Proximity

Refines position only when optional BLE beacons or UWB devices are in range.

Topological Graph Construction and Shortest Paths



Walk Capture

Loop Closure

Route Export

Converts user walking paths into connected network nodes (doors, turns, rooms) and edges (hallways). Detects loops and merges overlapping paths when matching magnetic signatures are re-encountered.

A* Graph Routing

- Calculates optimal hallway paths across floor nodes in under **15ms**.
- Supports accessible path filtering, dynamically removing stair edges for wheelchair-friendly navigation.

Data Portability

Exports floor networks directly as standard GeoJSON files for inspection and editing on geojson.io.

Floor Detection and Short-Range AR Anchoring

Vertical Transitions

01

Pressure Delta Monitoring

Monitors relative barometric pressure deltas. An altitude gain above **3.0m** triggers an automatic floor layer switch.

02

Transport Mode Discrimination

Distinguishes elevator rides (rapid pressure shift without footsteps) from stair climbing (rhythmic steps with 180-degree turns).

03

Manual Override

Includes an on-screen vertical floor rail for manual floor selection at any time.

AR Guidance Strategy

Short-Range Arrow Projection

Projects 3D directional arrows only **2.5 meters** ahead along the active hallway path, preventing long-distance camera drift.

Tilt-Based View Switching

Holding the phone upright activates the AR camera view; holding it flat switches to the 2D floor map to conserve battery.

Cloud Setup, Storage Pipeline, and Data Validation

1

Tier 1: Frontend

GitHub Pages hosting static HTML5 canvas and presentation views.

2

Tier 2: Compute Engine

PythonAnywhere running Python 3.10, Flask, and SQLite3 metadata storage.

3

Tier 3: Object Storage

Backblaze B2 bucket accessed via AWS Boto3 SDK using S3 Signature v4.

Data Integrity Safeguards

Speed Filter

Discards recorded traces with average walking speeds exceeding **2.5 m/s**.

Staging Buffer

Holds submitted user paths in a staging queue before public release.

DBSCAN Consensus

Requires **3 independent matching walks** within **1.5m** before adding a path to the live public graph.

Admin Locking

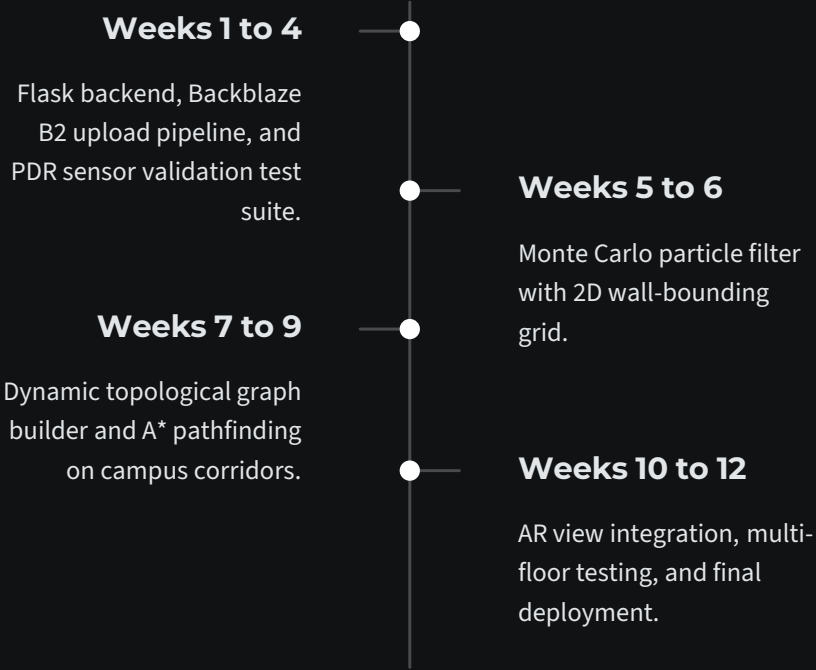
Allows verified floor maps to be locked against structural changes.

Risk Mitigation and Development Milestones

Technical Risk Matrix

Risk	Mitigation
Sensor Drift	Particle filter wall constraints and magnetic signature corrections.
Barometer Fluctuations	Tracking relative pressure changes instead of absolute values, backed by a manual UI override.

Milestone Timeline



Development Timeline and Milestones

